CYBERSECURITY LAB SETUP REPORT (TASK-2)

Title: Build Your Personal Cybersecurity Lab – Task 2
Name: Rosemary J
Course/Internship: Cybersecurity and Ethical Hacking Internship
Organization: Maincrafts Technology
Date: 25/02/2026

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1. Objective

The objective of this task is to build a safe and isolated cybersecurity lab environment using virtualization. The lab includes an attacker machine (Kali Linux), a vulnerable web application (DVWA), and security tools such as Burp Suite, Wireshark, and Nmap for practicing penetration testing and traffic analysis safely.

2. Tools and Software Used

Oracle VirtualBox:

Oracle VirtualBox is a virtualization software used to create and run virtual machines on a host system. It allows multiple operating systems to run simultaneously in an isolated environment, making it ideal for building a safe cybersecurity lab.

Kali Linux Virtual Machine:

Kali Linux is a penetration testing and ethical hacking operating system that comes pre-installed with various security tools. In this lab, it was used as the attacker machine to perform scanning, traffic analysis, and vulnerability testing.

DVWA (Damn Vulnerable Web Application):

DVWA is a deliberately vulnerable web application designed for practicing web application security testing. It contains common vulnerabilities such as SQL Injection, XSS, and File Inclusion, which help users understand real-world attack techniques in a controlled environment.

Burp Suite:

Burp Suite is a web application security testing tool used to intercept, inspect, and modify HTTP requests and responses. In this lab, it was used to configure a proxy and capture traffic between the browser and DVWA.

Wireshark Network Analyzer:

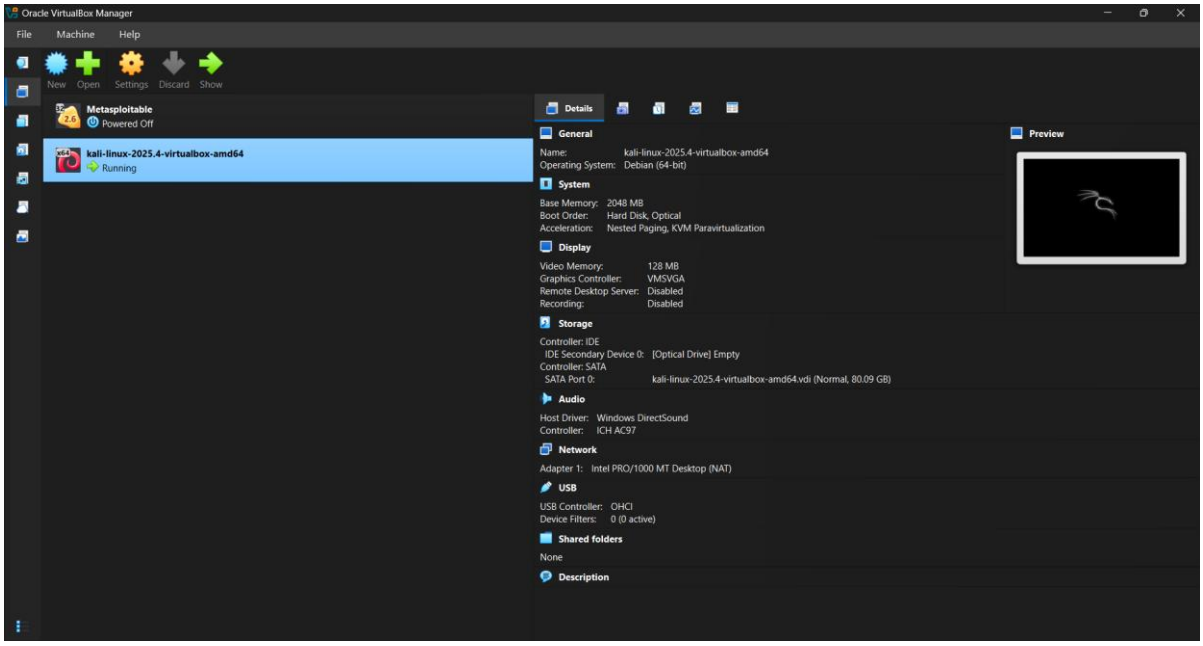
Wireshark is a network protocol analyzer used to capture and inspect network traffic in real time. It was used in this lab to monitor packets and analyze communication between the attacker machine and the target application.

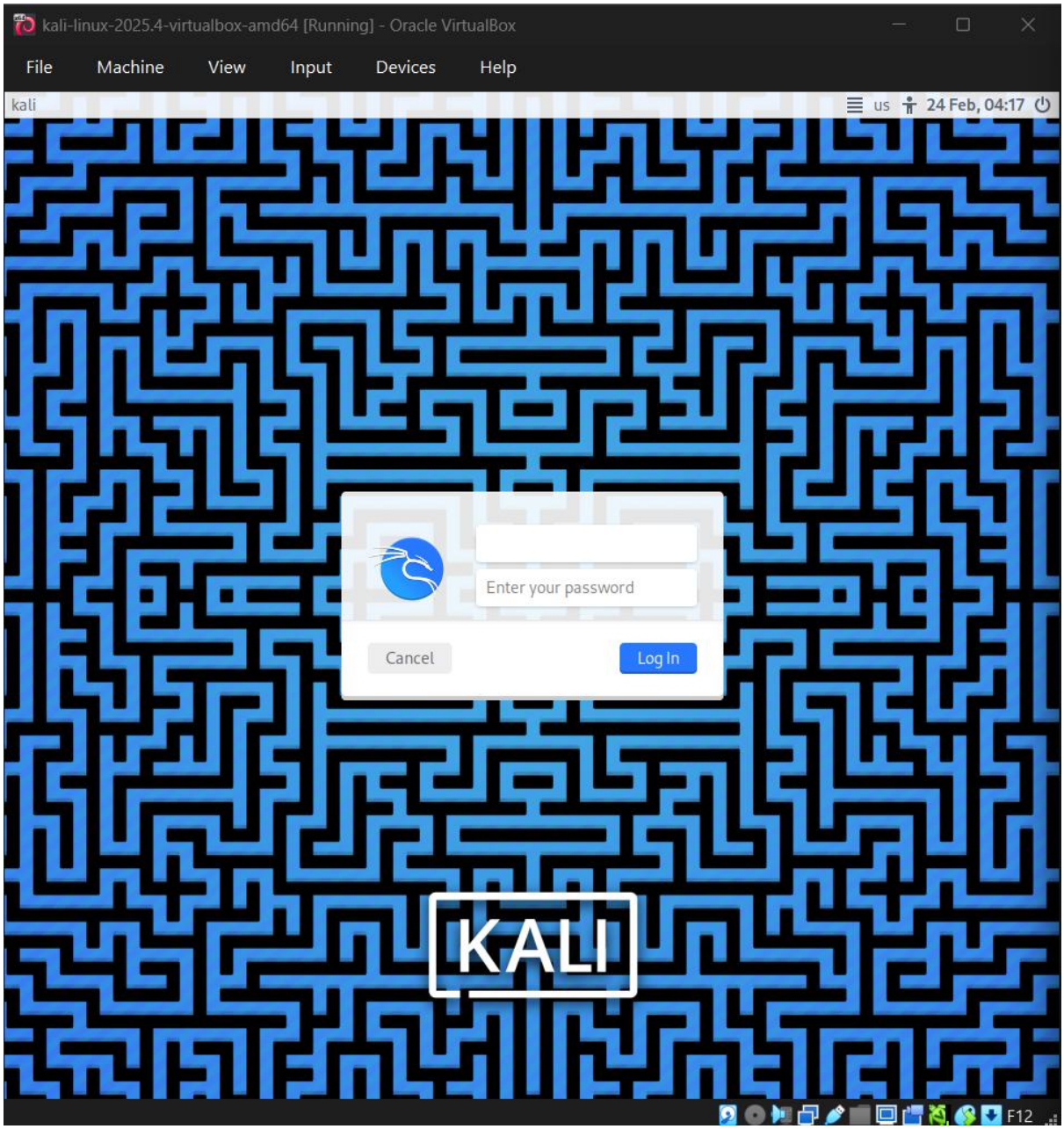
Nmap Scanner:

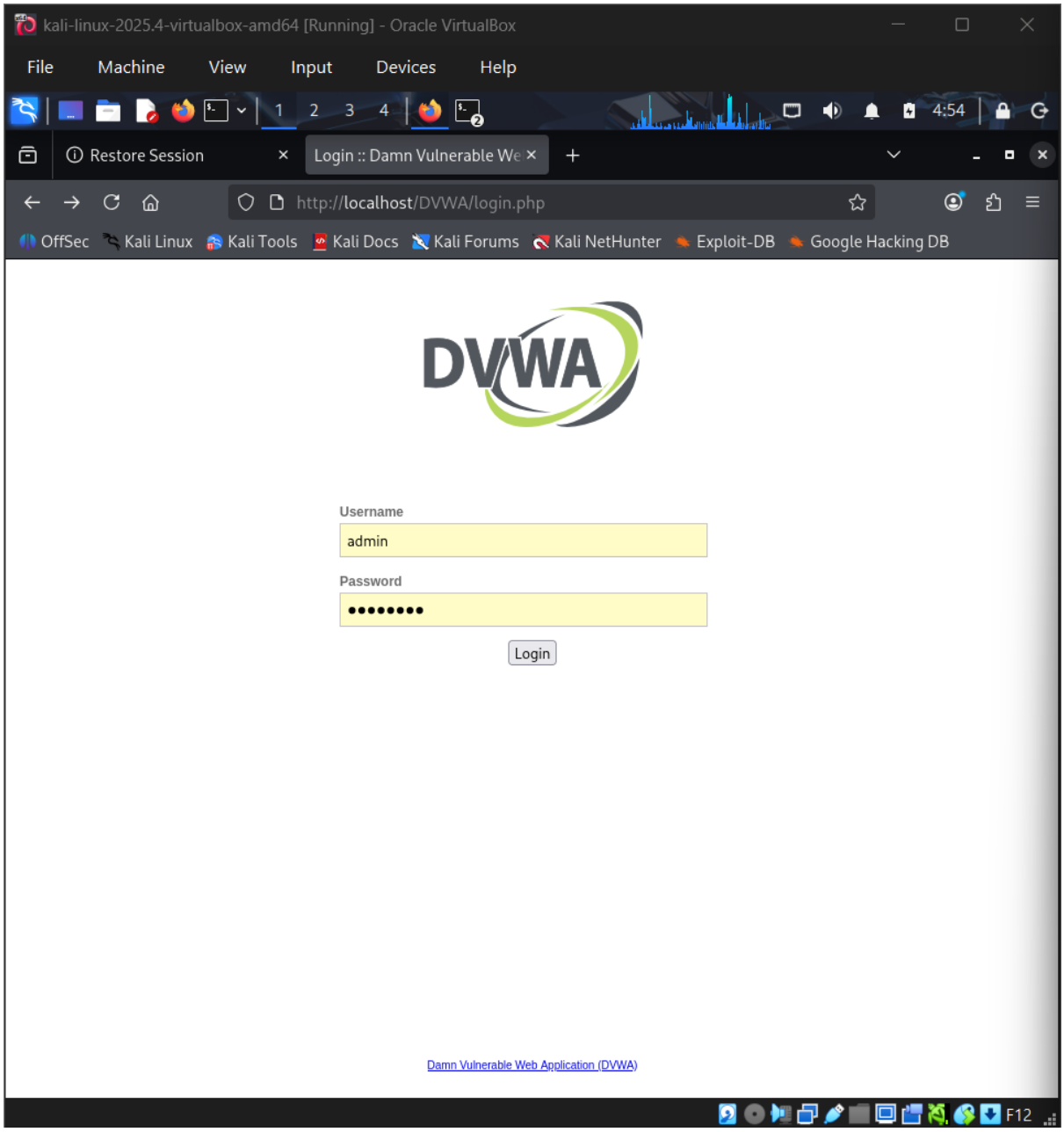
Nmap (Network Mapper) is a network scanning tool used to discover hosts, open ports, and running services. In this lab, it was used to identify active services on the DVWA server, such as the Apache web server running on port 80.

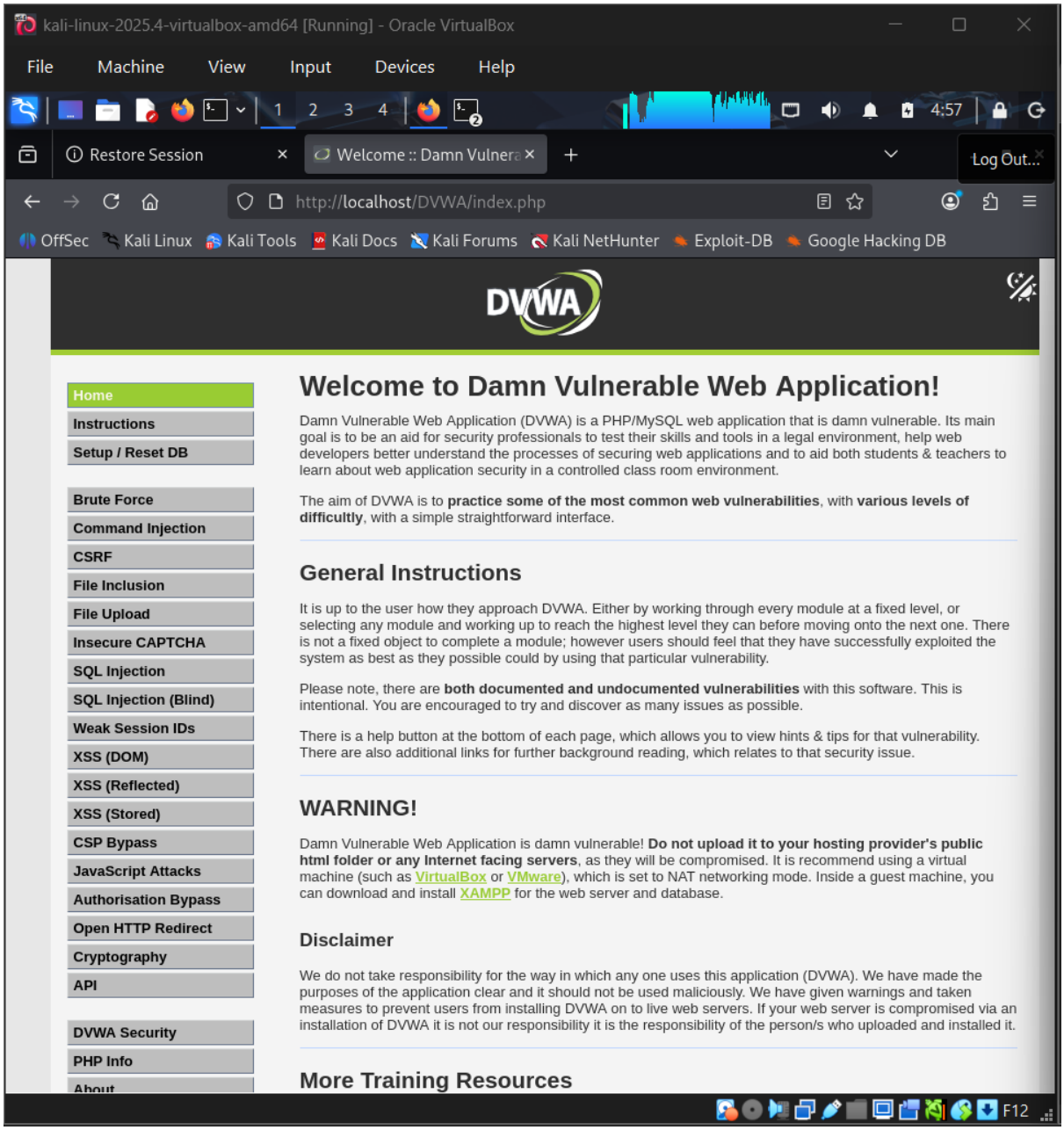
3. Lab Setup

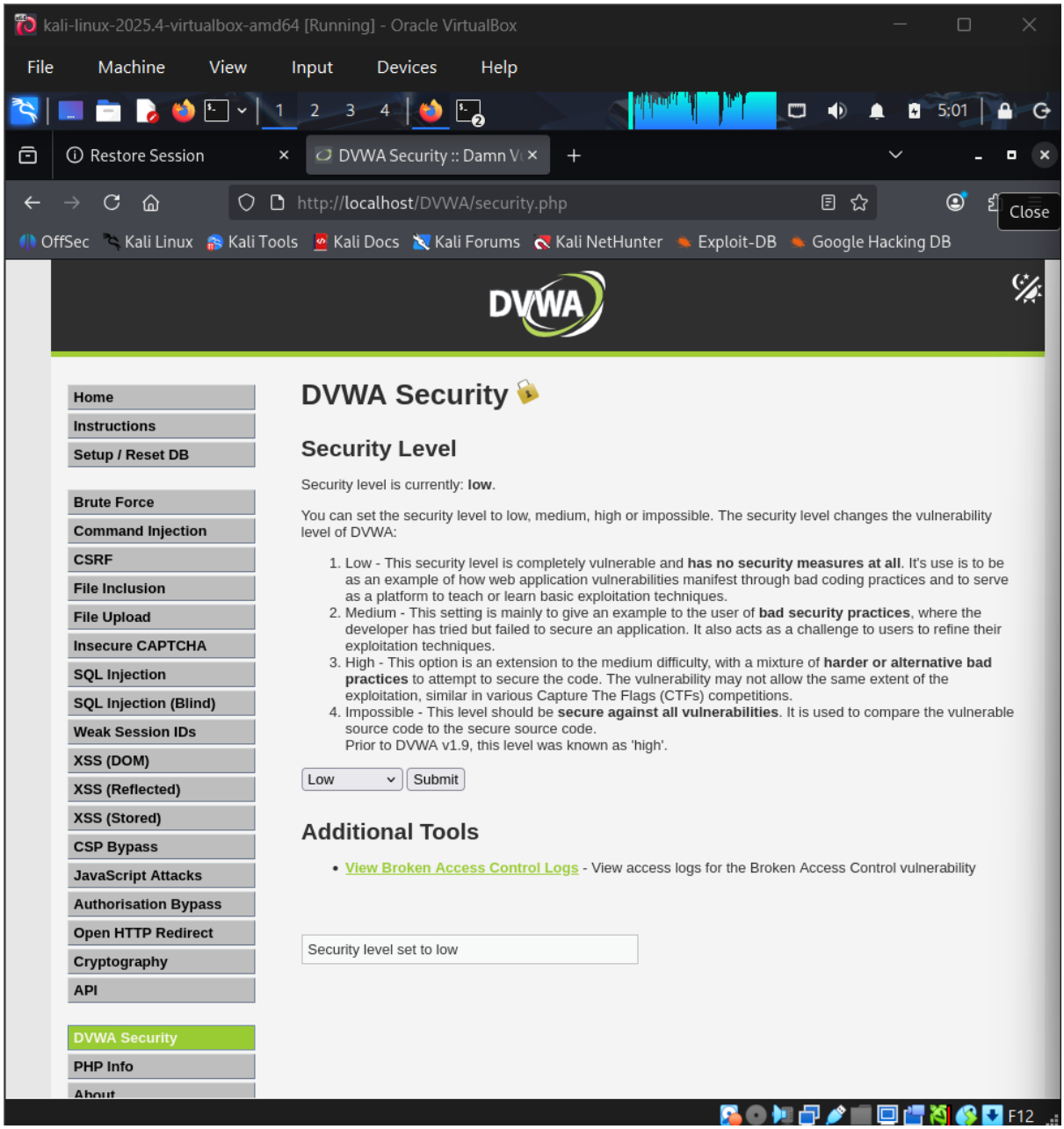
Kali Linux virtual machine was configured in VirtualBox with sufficient RAM, CPU, and storage allocation. DVWA was installed on the Apache web server running on Kali Linux, and the security level was set to low for testing purposes.




kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
kali
us
24 Feb, 04:17
Enter your password
Cancel
Log In
KALI
F12

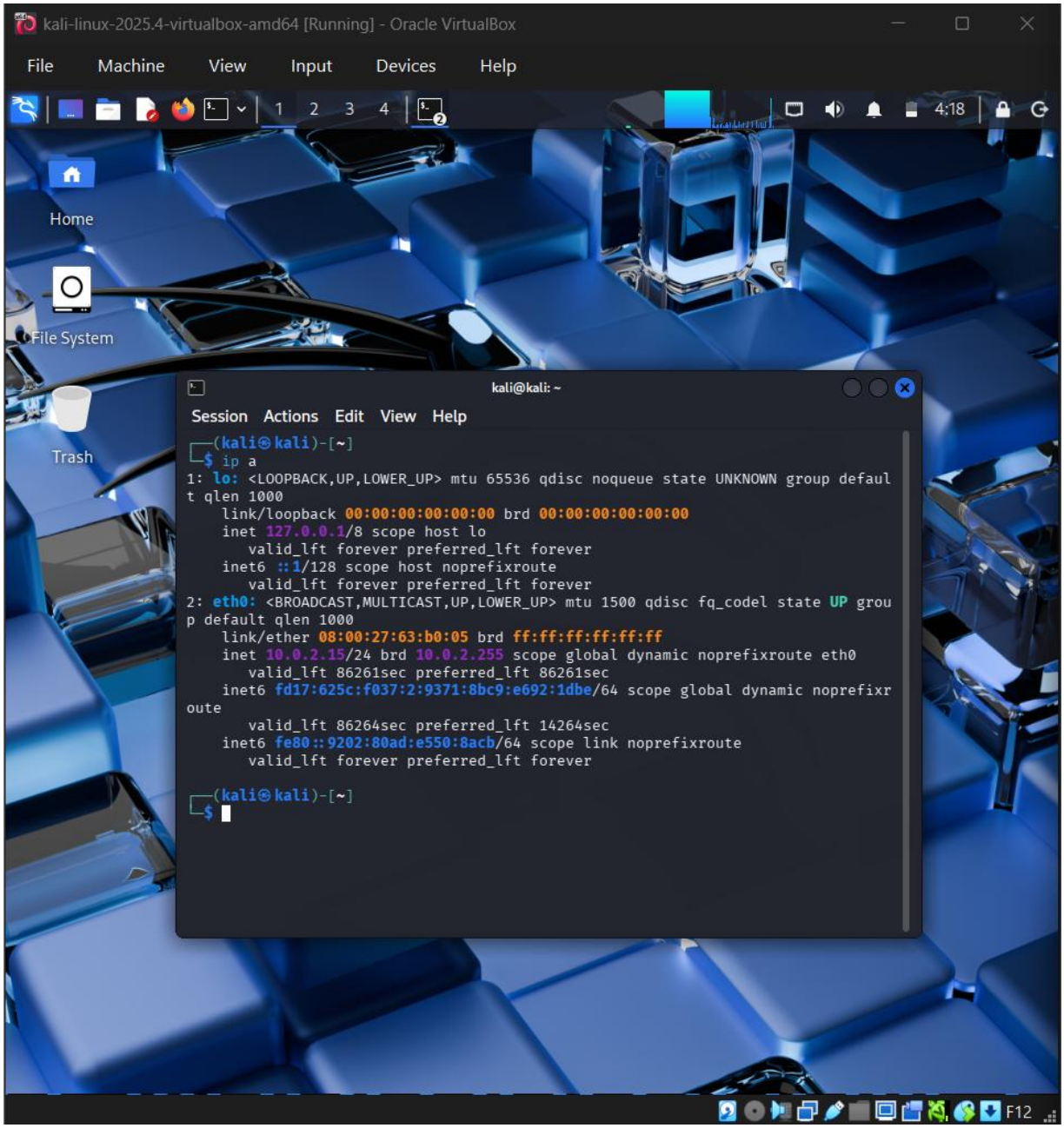

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Restore Session
Login :: Damn Vulnerable We
http://localhost/DVWA/login.php
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
DVWA
Username
admin
Password
Login
Damn Vulnerable Web Application (DVWA)

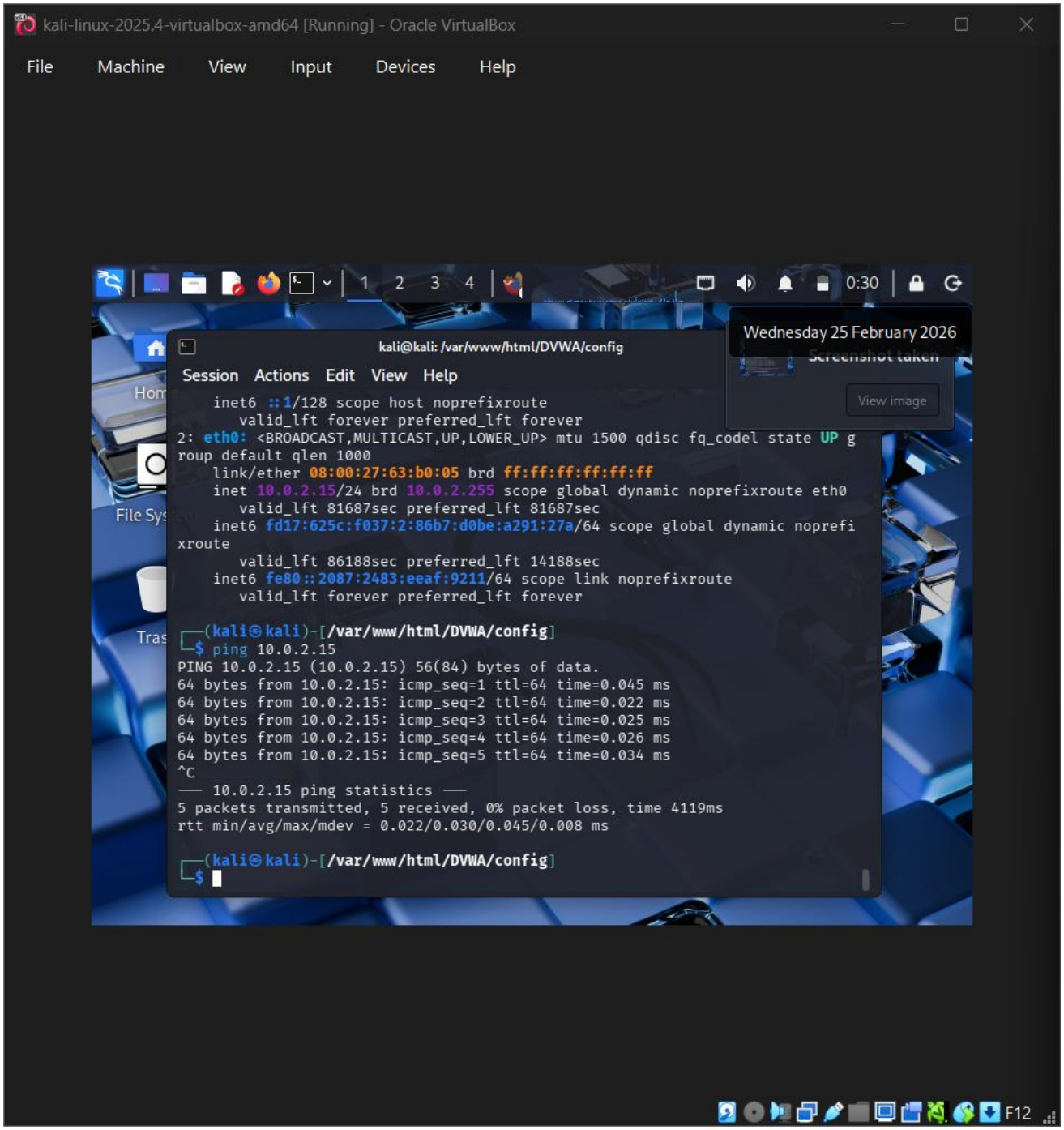

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Restore Session
Welcome :: Damn Vulnera
Log Out...
http://localhost/DVWA/index.php
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript Attacks
Authorisation Bypass
Open HTTP Redirect
Cryptography
API
DVWA Security
PHP Info
Welcome to Damn Vulnerable Web Application!
Damn Vulnerable Web Application (DVWA) is a PHP/MySQL web application that is damn vulnerable. Its main goal is to be an aid for security professionals to test their skills and tools in a legal environment, help web developers better understand the processes of securing web applications and to aid both students & teachers to learn about web application security in a controlled class room environment.
The aim of DVWA is to practice some of the most common web vulnerabilities, with various levels of difficultly, with a simple straightforward interface.
General Instructions
It is up to the user how they approach DVWA. Either by working through every module at a fixed level, or selecting any module and working up to reach the highest level they can before moving onto the next one. There is not a fixed object to complete a module; however users should feel that they have successfully exploited the system as best as they possible could by using that particular vulnerability.
Please note, there are both documented and undocumented vulnerabilities with this software. This is intentional. You are encouraged to try and discover as many issues as possible.
There is a help button at the bottom of each page, which allows you to view hints & tips for that vulnerability. There are also additional links for further background reading, which relates to that security issue.
WARNING!
Damn Vulnerable Web Application is damn vulnerable! Do not upload it to your hosting provider's public html folder or any Internet facing servers, as they will be compromised. It is recommend using a virtual machine (such as VirtualBox or VMware), which is set to NAT networking mode. Inside a guest machine, you can download and install XAMPP for the web server and database.
Disclaimer
We do not take responsibility for the way in which any one uses this application (DVWA). We have made the purposes of the application clear and it should not be used maliciously. We have given warnings and taken measures to prevent users from installing DVWA on to live web servers. If your web server is compromised via an installation of DVWA it is not our responsibility it is the responsibility of the person/s who uploaded and installed it.
More Training Resources


kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Restore Session
DVWA Security :: Damn V
http://localhost/DVWA/security.php
Close
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript Attacks
Authorisation Bypass
Open HTTP Redirect
Cryptography
API
DVWA Security
PHP Info
About
DVWA Security
Security Level
Security level is currently: low.
You can set the security level to low, medium, high or impossible. The security level changes the vulnerability level of DVWA:
1. Low - This security level is completely vulnerable and has no security measures at all. It's use is to be as an example of how web application vulnerabilities manifest through bad coding practices and to serve as a platform to teach or learn basic exploitation techniques.
2. Medium - This setting is mainly to give an example to the user of bad security practices, where the developer has tried but failed to secure an application. It also acts as a challenge to users to refine their exploitation techniques.
3. High - This option is an extension to the medium difficulty, with a mixture of harder or alternative bad practices to attempt to secure the code. The vulnerability may not allow the same extent of the exploitation, similar in various Capture The Flags (CTFs) competitions.
4. Impossible - This level should be secure against all vulnerabilities. It is used to compare the vulnerable source code to the secure source code.
Prior to DVWA v1.9, this level was known as 'high'.
Low
Submit
Additional Tools
View Broken Access Control Logs - View access logs for the Broken Access Control vulnerability
Security level set to low

4. Network Configuration

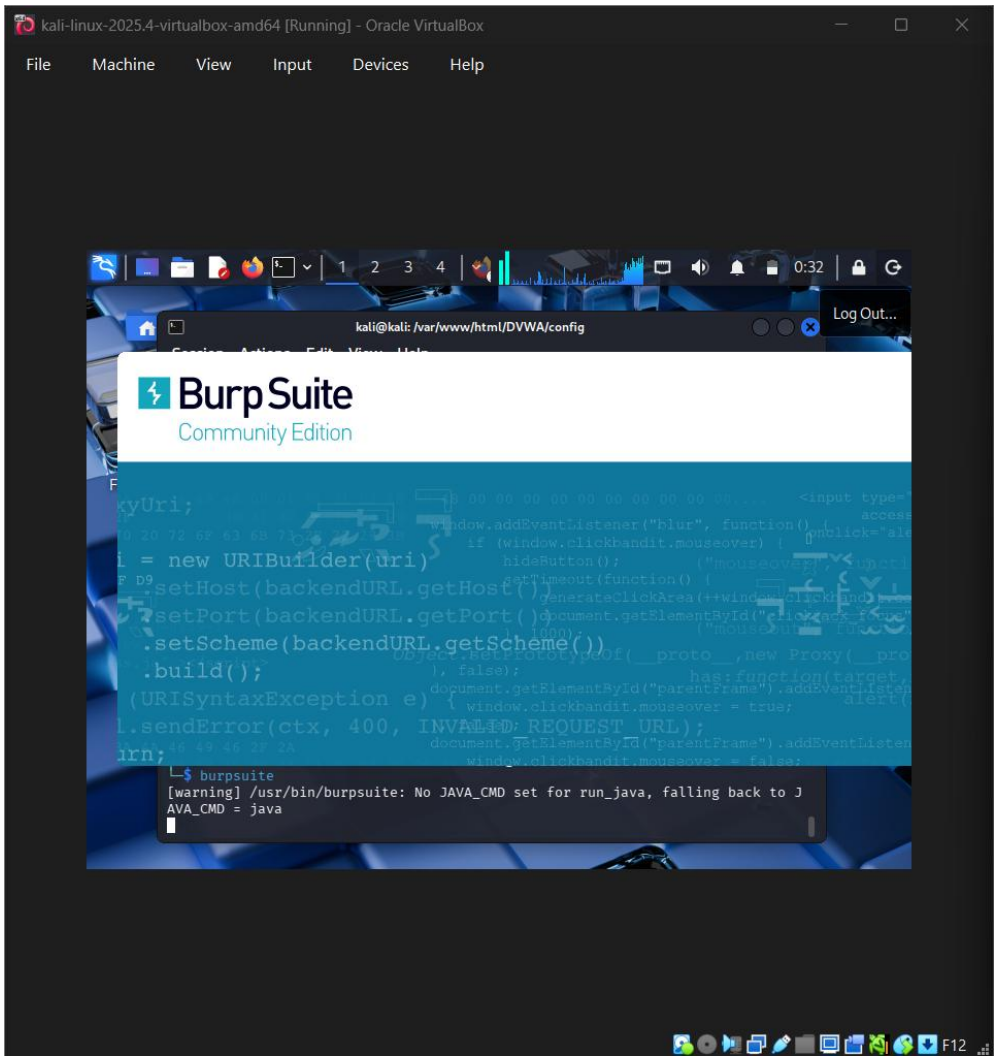
Network connectivity was verified using IP configuration commands. Ping tests confirmed proper communication between systems and the DVWA server.

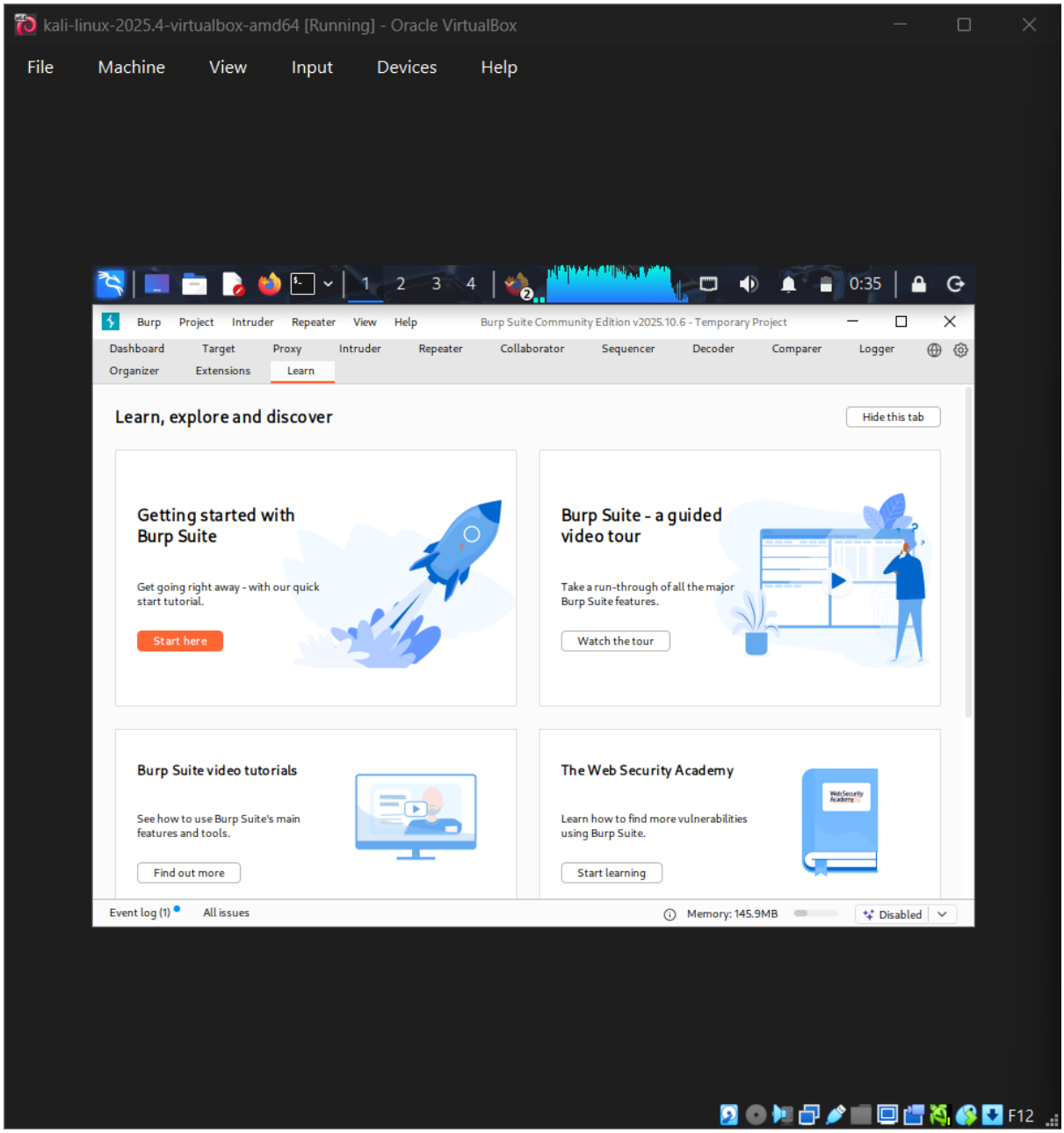


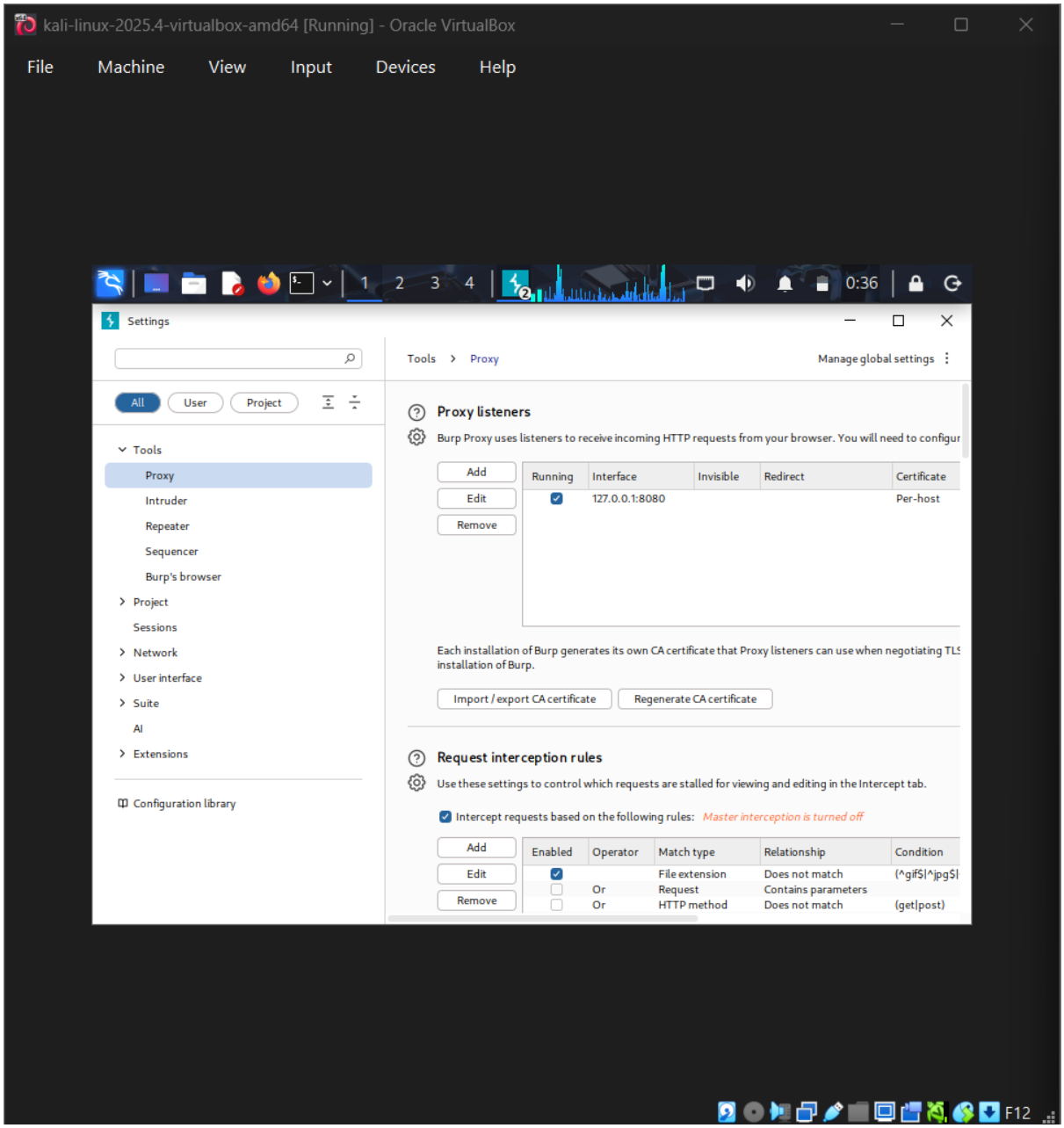

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: /var/www/html/DVWA/config
Session Actions Edit View Help
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
valid_lft 81687sec preferred_lft 81687sec
inet6 fd17:625c:f037:2:86b7:d0be:a291:27a/64 scope global dynamic noprefixroute
valid_lft 86188sec preferred_lft 14188sec
inet6 fe80::2087:2483:eeaf:9211/64 scope link noprefixroute
valid_lft forever preferred_lft forever
(kali㉿kali)-[/var/www/html/DVWA/config]
$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.045 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.022 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.025 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.026 ms
64 bytes from 10.0.2.15: icmp_seq=5 ttl=64 time=0.034 ms
^C
— 10.0.2.15 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4119ms
rtt min/avg/max/mdev = 0.022/0.030/0.045/0.008 ms
(kali㉿kali)-[/var/www/html/DVWA/config]
$
Wednesday 25 February 2026
View image
0:30
F12

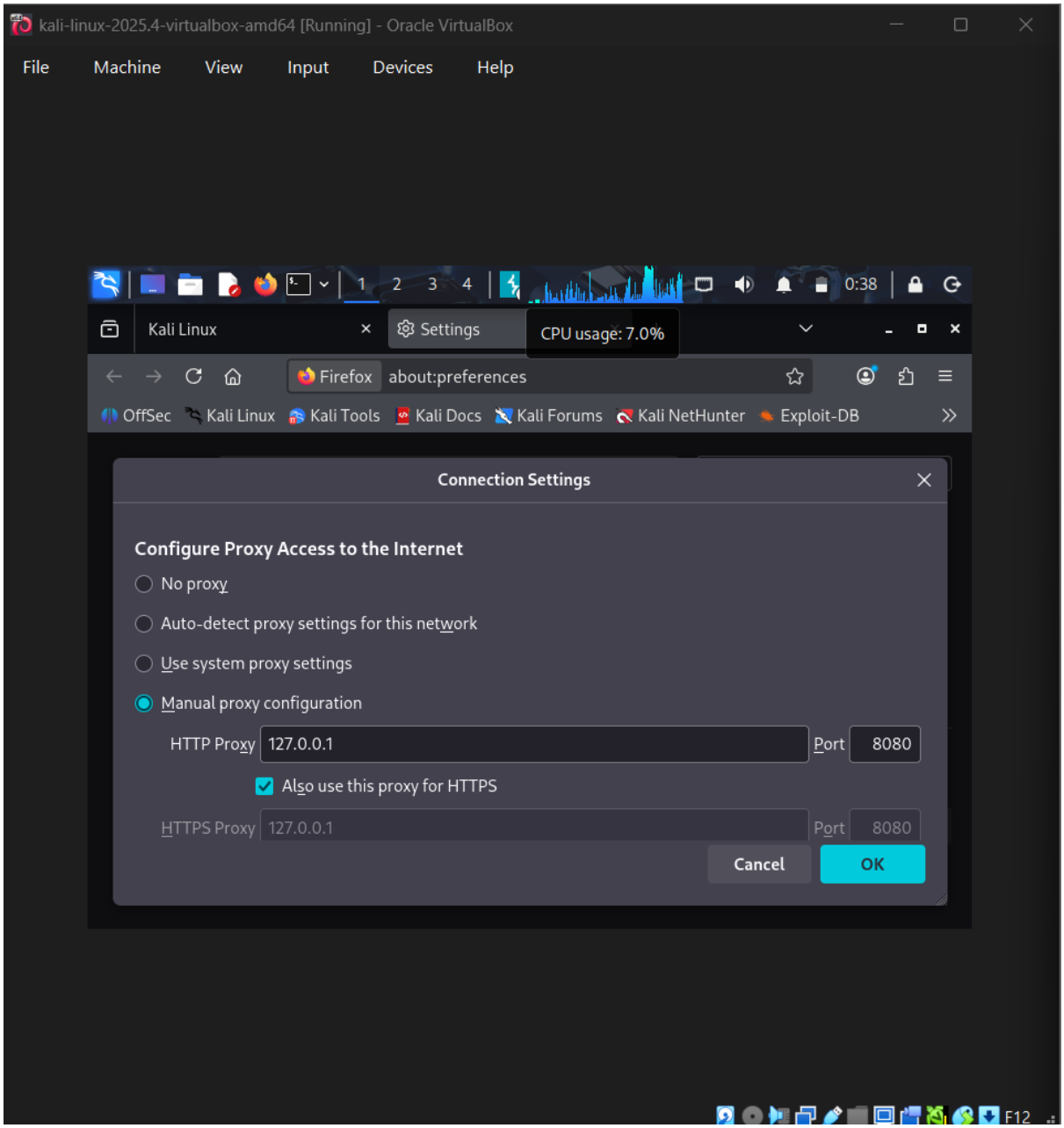
5. Burp Suite Validation

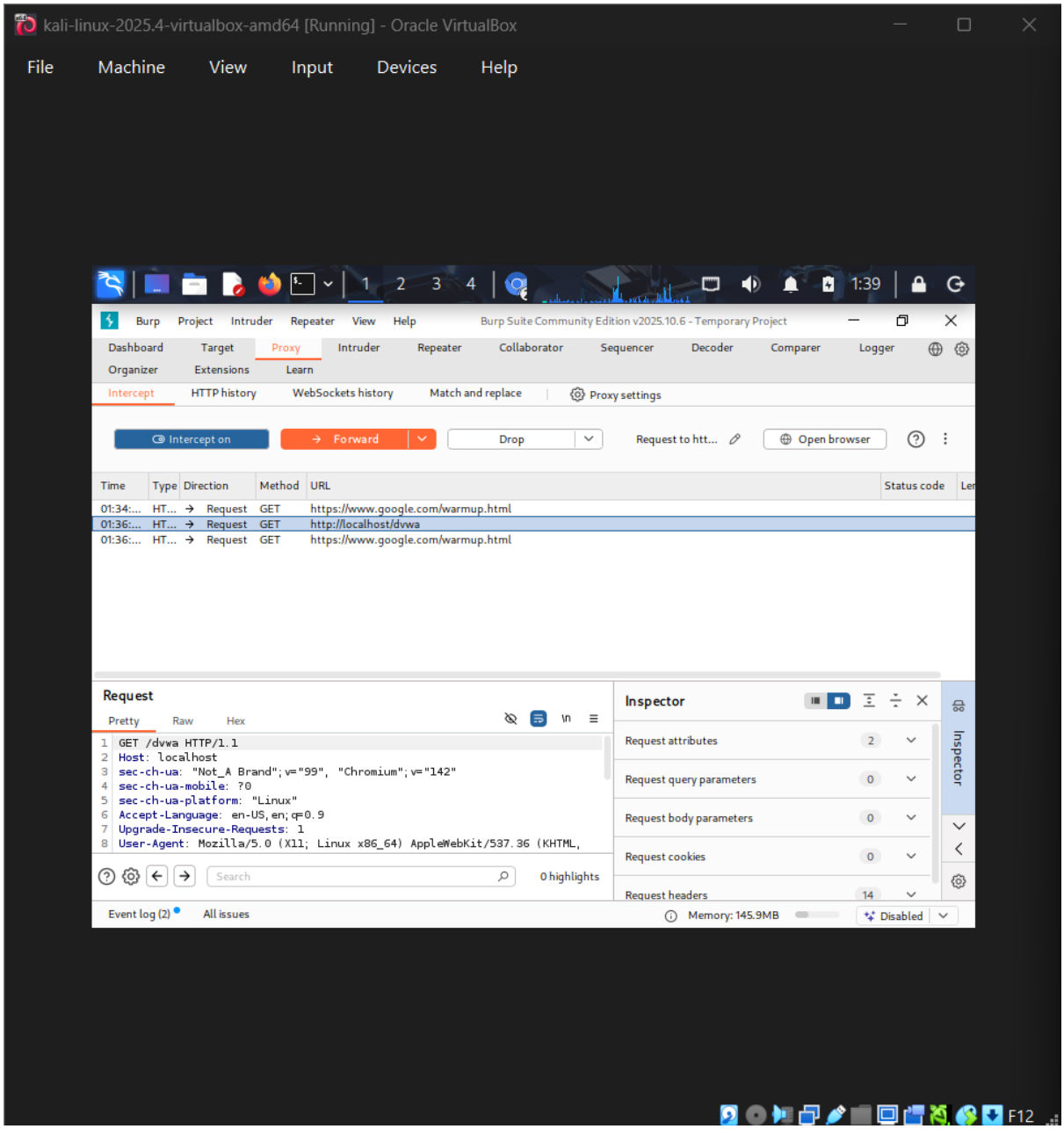
Burp Suite proxy listener was configured on 127.0.0.1:8080. Firefox proxy settings were configured accordingly. HTTP requests from DVWA were successfully intercepted and forwarded, confirming proper proxy functionality.




kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Burp
Project
Intruder
Repeater
View
Help
Burp Suite Community Edition v2025.10.6 - Temporary Project
Dashboard
Target
Proxy
Intruder
Repeater
Collaborator
Sequencer
Decoder
Comparer
Logger
Organizer
Extensions
Learn
Learn, explore and discover
Hide this tab
Getting started with Burp Suite
Get going right away - with our quick start tutorial.
Start here
Burp Suite - a guided video tour
Take a run-through of all the major Burp Suite features.
Watch the tour
Burp Suite video tutorials
See how to use Burp Suite's main features and tools.
Find out more
The Web Security Academy
Learn how to find more vulnerabilities using Burp Suite.
Start learning
Event log (1)
All issues
Memory: 145.9MB
Disabled
F12

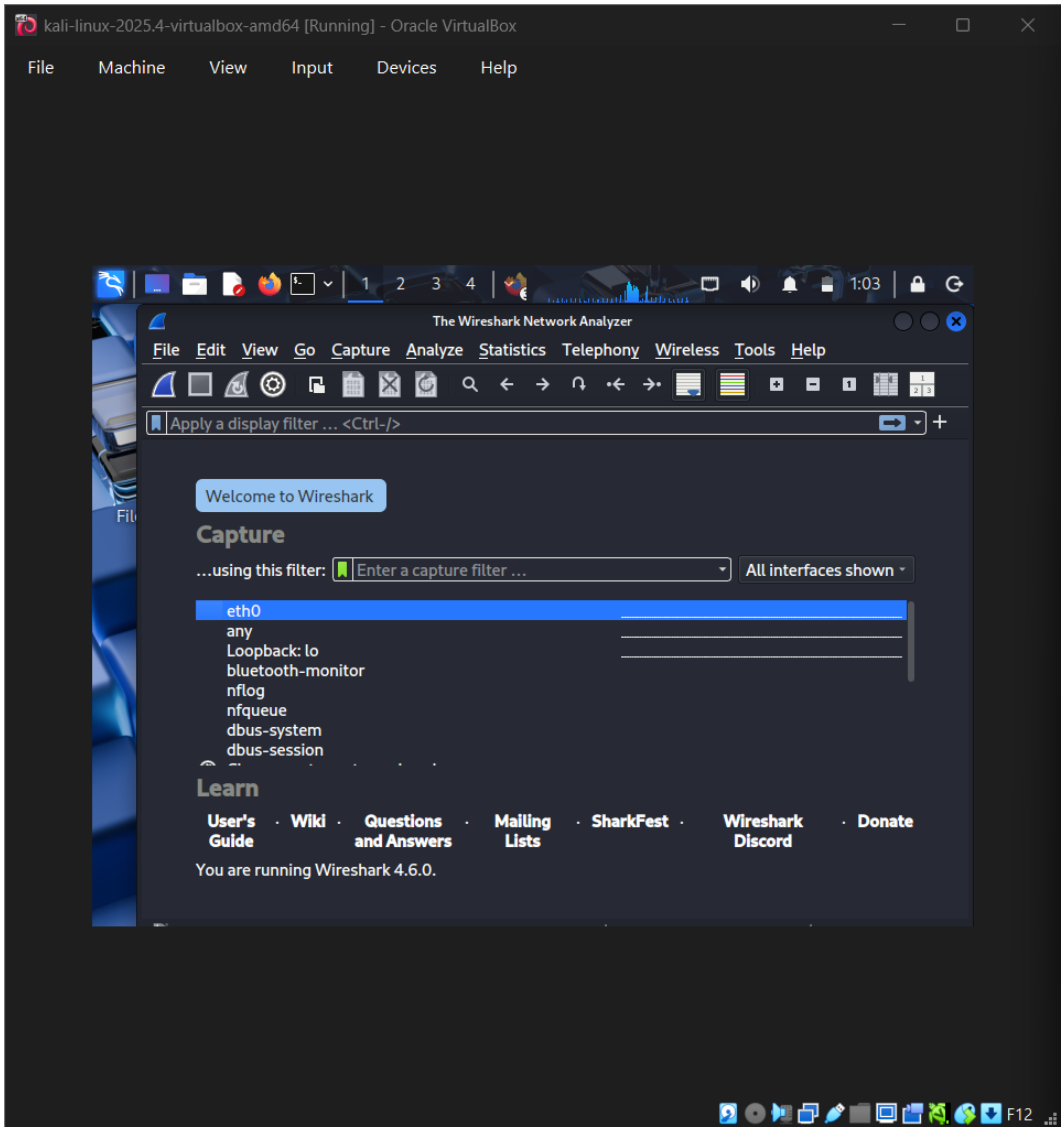

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Settings
Tools > Proxy
Manage global settings
All
User
Project
Tools
Proxy
Intruder
Repeater
Sequencer
Burp's browser
Project
Sessions
Network
User interface
Suite
AI
Extensions
Configuration library
Proxy listeners
Burp Proxy uses listeners to receive incoming HTTP requests from your browser. You will need to configur
Add
Edit
Remove
Running
Interface
Invisible
Redirect
Certificate
127.0.0.1:8080
Per-host
Each installation of Burp generates its own CA certificate that Proxy listeners can use when negotiating TLS
installation of Burp.
Import / export CA certificate
Regenerate CA certificate
Request interception rules
Use these settings to control which requests are stalled for viewing and editing in the Intercept tab.
Intercept requests based on the following rules:
Master interception is turned off
Enabled
Operator
Match type
Relationship
Condition
File extension
Does not match
(^gif$|^jpg$|
Or
Request
Contains parameters
Or
HTTP method
Does not match
(get|post)

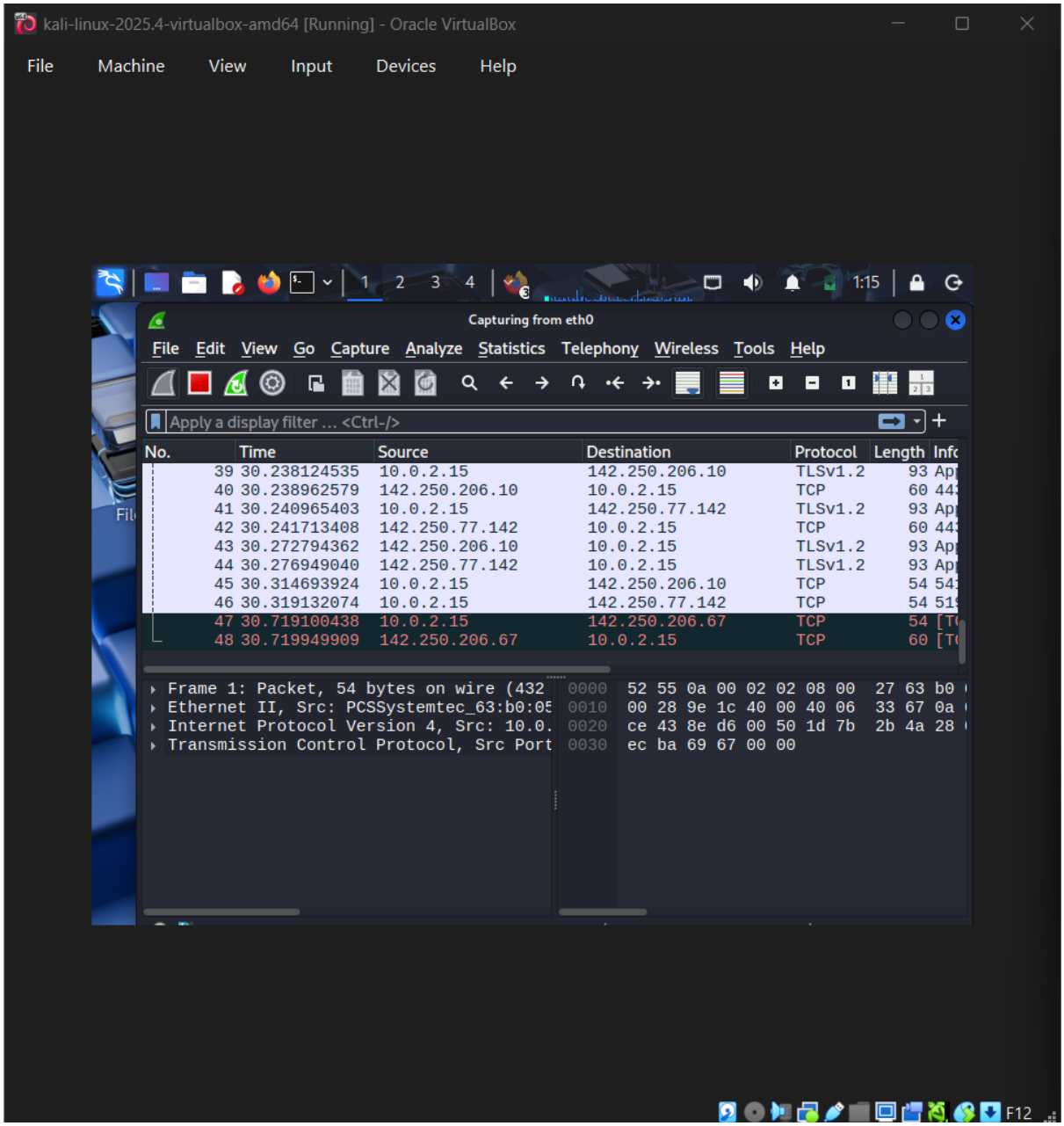

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Kali Linux
Settings
CPU usage: 7.0%
Firefox
about:preferences
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Connection Settings
Configure Proxy Access to the Internet
No proxy
Auto-detect proxy settings for this network
Use system proxy settings
Manual proxy configuration
HTTP Proxy
127.0.0.1
Port
8080
Also use this proxy for HTTPS
HTTPS Proxy
127.0.0.1
Port
8080
Cancel
OK


kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Burp Project Intruder Repeater View Help
Burp Suite Community Edition v2025.10.6 - Temporary Project
Dashboard Target Proxy Intruder Repeater Collaborator Sequencer Decoder Comparer Logger
Organizer Extensions Learn
Intercept HTTP history WebSockets history Match and replace Proxy settings
Intercept on Forward Drop Request to htt... Open browser
Time Type Direction Method URL Status code
01:34:... HT... Request GET https://www.google.com/warmup.html
01:36:... HT... Request GET http://localhost/dvwa
01:36:... HT... Request GET https://www.google.com/warmup.html
Request
Pretty Raw Hex
GET /dvwa HTTP/1.1
Host: localhost
sec-ch-ua: "Not_A Brand";v="99", "Chromium";v="142"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "Linux"
Accept-Language: en-US,en;q=0.9
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML,
Search 0 highlights
Inspector
Request attributes 2
Request query parameters 0
Request body parameters 0
Request cookies 0
Request headers 14
Event log (2) All issues Memory: 145.9MB Disabled

6. Wireshark Traffic Analysis

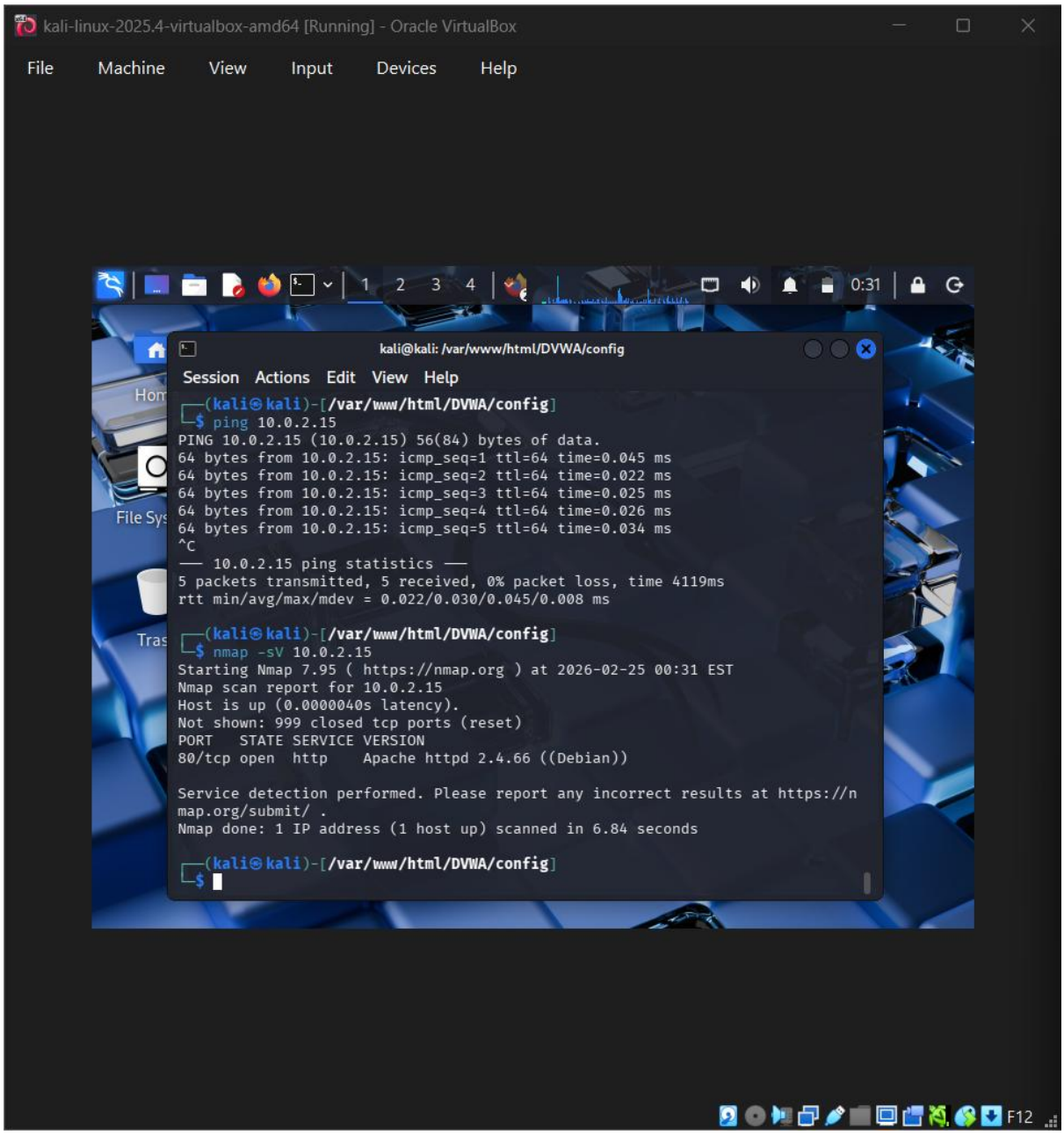
Wireshark was used to capture network packets. The eth0 interface was selected and DVWA traffic was captured successfully for analysis.




kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
39 30.238124535 10.0.2.15 142.250.206.10 TLSv1.2 93
40 30.238962579 142.250.206.10 10.0.2.15 TCP 60
41 30.240965403 10.0.2.15 142.250.77.142 TLSv1.2 93
42 30.241713408 142.250.77.142 10.0.2.15 TCP 60
43 30.272794362 142.250.206.10 10.0.2.15 TLSv1.2 93
44 30.276949040 142.250.77.142 10.0.2.15 TLSv1.2 93
45 30.314693924 10.0.2.15 142.250.206.10 TCP 54
46 30.319132074 10.0.2.15 142.250.77.142 TCP 54
47 30.719100438 10.0.2.15 142.250.206.67 TCP 54
48 30.719949909 142.250.206.67 10.0.2.15 TCP 60
Frame 1: Packet, 54 bytes on wire (432
Ethernet II, Src: PCSSystemtec_63:b0:05
Internet Protocol Version 4, Src: 10.0.
Transmission Control Protocol, Src Port
0000 52 55 0a 00 02 02 08 00 27 63 b0
0010 00 28 9e 1c 40 00 40 06 33 67 0a
0020 ce 43 8e d6 00 50 1d 7b 2b 4a 28
0030 ec ba 69 67 00 00
F12

7. Vulnerability Scanning Using Nmap

An Nmap scan was performed on the DVWA host to identify open ports and services. The Apache HTTP service was detected on port 80, confirming the presence of the web application.



8. Results

The cybersecurity lab environment was successfully created. Kali Linux attacker machine communicated with DVWA target application. Security tools such as Burp Suite, Wireshark, and Nmap were successfully tested.

9. Conclusion

The cybersecurity lab setup was successfully completed using Kali Linux and DVWA. The configured environment allows safe practice of penetration testing, traffic analysis, and vulnerability scanning in an isolated lab.